

A Basis for the Dyadic Sections of Euler's Totient

A structure theorem, a sharp Farey denominator bound, and the exact boundary of a kernel-checked attack on Erdős #249

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ABSTRACT

Erdős #249 asks whether $S = \sum_{n \geq 1} \varphi(n)/2^n$ is irrational. It is open, nothing here closes it, and this note reports what a kernel-checked attack does yield, each claim bounded by the exact proposition a proof checker accepted — the negative results included.

The 2-kernel of a sequence c is the family of subsequences $n \mapsto c(2^j n + r)$ for $j \geq 0$ and $0 \leq r < 2^j$. Coons proved that φ is not k -regular for any $k \geq 2$, that is, that the $\mathbb{Q}$ -span of its k -kernel is not finitely generated; Bell and Smertnig strengthened this to non-Mahlerity. Both are statements about the union over all levels. Our main theorem is the complementary positive one, at each finite level: writing $\varphi_{j,r}: n \mapsto \varphi(2^j n + r)$, the sections $\varphi_{0,0}$, $\varphi_{1,0}$ and $\varphi_{j,r}$ with r odd form a $\mathbb{Q}$ -basis of the span of *all* the $\varphi_{j,r}$; the span through level e has dimension exactly $2^e + 1$ for every $e \geq 1$; and the only $\mathbb{Q}$ -linear relations among dyadic sections of φ are those generated by two elementary identities.

Transporting that rank through binary long division gives the one consequence for S we can state: rationality would force a tempered integral carry whose sections through level e span at least $2^e - 1$ dimensions, for every e . This is an anti-compression barrier and not a criterion, because no finite-rank upper bound is proved for the carries rationality supplies.

We also record the exact reformulations, four equivalences and one implication, kept separate because an equivalence is not progress; a Farey denominator exclusion below 7.96×10^{34} , whose method is classical and whose scale is a function of how many binary digits were committed; and the proof that no fixed positive integer clears the coordinates $A(n)/n$ of the Lambert weight $A = \varphi * \mu$.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

1 Results

Every item is a proposition the pinned Lean kernel accepted, or (R10) the problem itself; the tag gives its status in the claim registry's taxonomy, and each *Checked* line links the declarations at commit 571ec44f2aad. Nothing in the list is an irrationality criterion for S , and no combination of the items becomes one. Throughout $\varphi_{j,r}(n) = \varphi(2^j n + r)$, for $j \geq 0$ and $0 \leq r < 2^j$, is the (j, r) *dyadic section* of Euler's totient, a vector in $\mathbb{Q}^{\mathbb{N}}$. Two elementary identities relate them: the *reductions* $\varphi_{j+1,0} = 2^j \varphi_{1,0}$ ([checked](#)) and $\varphi_{j,2^{t+1}s} = 2^t \varphi_{j-t-1,s}$ for s odd with $2^{t+1}s < 2^j$ ([checked](#)).

R1. [PROVED HERE] **Basis theorem.** The family $\mathcal{B} = \{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, r \text{ odd}, 0 < r < 2^j\}$ is linearly independent over $\mathbb{Q}$ and spans the same $\mathbb{Q}$ -subspace of $\mathbb{Q}^{\mathbb{N}}$ as the set of *all* dyadic sections $\{\varphi_{j,r} : j \geq 0, 0 \leq r < 2^j\}$; so $\mathcal{B}$ is a basis of that span, which is therefore infinite-dimensional. Since the reductions make every remaining section a rational multiple of a member of $\mathcal{B}$, the $\mathbb{Q}$ -linear relations among the dyadic sections of φ are exactly those the reductions generate — a one-line consequence of the checked statements, not a further one. *Checked:* [independence](#), [span equality](#), [basis](#).

R2. [PROVED HERE] **Exact finite-level rank.** For every $e \geq 0$ the $2^e + 1$ sections $\varphi_{0,0}, \varphi_{1,0}$ and $\varphi_{j,r}$ with $1 \leq j \leq e, r \text{ odd}, 0 < r < 2^j$ are linearly independent over $\mathbb{Q}$, so their span has dimension exactly $2^e + 1 = 2 + \sum_{j=1}^e 2^{j-1}$: exact, not an estimate, because the even residues are already dependent by the reductions. Read instead as a truncation of the kernel, $V_e = \text{span}_{\mathbb{Q}}\{\varphi_{j,r} : 0 \leq j \leq e\}$ has $\dim V_e = 2^e + 1$ for every $e \geq 1$, and $\dim V_0 = 1$: the counted family carries $\varphi_{1,0}$, which is a level-one section, so the two readings agree from $e = 1$ on and differ only at $e = 0$. The independence half of R1 follows by finite character; the span equality needs the reductions as well. *Checked:* [independence](#), [dimension](#).

Taxonomy adjudication for R1–R2. The registry groups these declarations with broader certificate and infinite-dimensional consequences under one coarse *unconditional progress* claim. The two rows above deliberately use *proved here*: for the narrower basis and exact-rank propositions this records kernel acceptance and makes no novelty assertion. Promoting either split row would require a separate prior-art judgement, not merely a declaration link.

R3. [FORMALISED HERE] **Denominator exclusion, sharp for its window.** If $S = a/q$ with $a \in \mathbb{Z}$ and $q \geq 1$ then $q > 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$. The constant is optimal for the window producing it: the next integer up is the exact first denominator at which that window's certificate fails. The method is Farey (Section 4). Here *formalised here* is the registry's evidence-status label: known mathematics represented and checked in Lean, not a novelty claim; the method is classical. *Checked:* [exclusion](#), [reduced-denominator form](#), [first failure](#).

R4. [PROVED HERE] **Carry anti-compression.** If S were rational then some tempered integral carry orbit for the coefficients φ , with positive multiplier, would have for *every* e carry sections through level e spanning a space of dimension at least $2^e - 1$. *Closes:* every rationality proof that would exhibit a carry of bounded section rank. *Does not close:* anything else — it is not an irrationality criterion, because no finite-rank upper bound is proved for the carries that rationality supplies. The proposition is unconditional in its hypotheses, but its proof is routine transport from R2 and has no separate prior-art assessment; *proved here* therefore records kernel acceptance without making the stronger contribution judgement *unconditional progress*. *Checked:* [the barrier](#), [rank transport from R2](#).

- R5.** [FORMALISED HERE] **The #249 weight.** $S = \sum_{d \geq 1} A(d)/(2^d - 1)$ with $A = \varphi * \mu$; here $A \geq 0$, $A(p) = p - 2$, $A(p^k) = \varphi(p^k) - \varphi(p^{k-1})$, and A is unbounded. *Checked:* [identity](#), [sign](#), [primes](#), [prime powers](#), [unboundedness](#).
- R6.** [FORMALISED HERE] **Eventually periodic weights are settled.** For every integer base $b \geq 2$, if $\gamma: \mathbb{N} \rightarrow \mathbb{Q}$ is eventually periodic, nonnegative and positive at some positive index of its periodic part, then $\sum_a \gamma(a)/(b^a - 1)$ is irrational (Luca and Tachiya prove a broader periodic theorem by other methods [6]; no priority is claimed). By **R5** the weight A is unbounded, hence not eventually periodic, so #249 lies just outside this class. *Checked:* [the periodic theorem](#).
- R7.** [PROVED HERE] **No fixed index clears the normalised weight.** For every integer $D \geq 1$ some $n \geq 1$ has $D \cdot A(n)/n \notin \mathbb{Z}$, and any D clearing every coordinate up to a horizon $N \geq 4$ is divisible by an explicit two-tier primorial (4 at $p = 2$, p^2 when $p^2 \leq N$, else p). *Closes:* every argument clearing the coordinates $A(n)/n$ by a single positive integer valid at all n . *Does not close:* arguments whose index grows with n , or which never clear the coordinates. *Checked:* [no fixed index](#), [the primorial divisibility](#).
- R8.** [PROVED HERE] **Four exact reformulations, offered as nothing more.** Irrationality of S is equivalent to each of: pointwise non-integrality of every tail difference; a cofinal supply of finite certificates at every fixed shift; and the same supply restricted to the lcm diagonal. A fourth equivalence, independent of the certificate language, is a counted anti-concentration condition on window phases. All four are restatements of #249 in other coordinates, and are the part of this work most likely to be mistaken for a result. *Checked:* [pointwise](#), [cofinal at each shift](#), [lcm diagonal](#), [window-separated pairs](#).
- R9.** [CONDITIONAL REDUCTION] **One implication, not an equivalence.** A supply of gap certificates — one window (N, K) per denominator q whose totient carry residue avoids a band of width $q(N+K+2)$ out of 2^K — *implies* irrationality. The converse is not proved and is not claimed; this is why the item is separate from **R8**. *Checked:* [the supply implication](#). *Registry:* no row currently owns this declaration.
- R10.** [OPEN] **Erdős #249.** Whether S is irrational. Not proved here.

2 Where #249 sits

If $f = g * \mathbf{1}$, that is $f(n) = \sum_{d|n} g(d)$, then interchanging the two sums gives

$$\sum_{n \geq 1} \frac{f(n)}{2^n} = \sum_{d \geq 1} \frac{g(d)}{2^d - 1}. \quad (1)$$

Four classical coefficient sequences have this shape. The table records the constant, the weight $g = f * \mu$, and its status. Decimals were recomputed to 50 places for this note and are truncated, not rounded.

The first three weights are bounded, taking only the values 0 and 1 (for τ the weight is the constant 1, so only the value 1). The fourth is unbounded, by **R5**. That difference is the one that matters for the route in **R6**, and we do not claim it is the only difference: A also vanishes on $n \equiv 2 \pmod{4}$, while $\mathbf{1}_{\text{primes}}$ is bounded but no more eventually periodic than A is, so the settled ω and Ω cases are not instances of **R6** either. They were settled by a different method, discussed in Section 9.

The difficulty of #249 does not live in the Lambert transform. The neighbouring rungs are rational and exactly computable — $\sum_{d \geq 1} \mu(d)/(2^d - 1) = \frac{1}{2}$ ([checked](#)) and

f	$\sum f(n)/2^n$	weight $g = f * \mu$	status
τ	$1.60669515 \dots = E$	$g \equiv 1$	Irrational (Erdős 1948); formalised here in the Lambert form $\sum_{d \geq 1} (2^d - 1)^{-1}$ (checked), equal to $\sum \tau(n)/2^n$ by (1)
ω	$0.51694281 \dots$	$g = \mathbf{1}_{\text{primes}}$	Irrational (Tao–Teräväinen 2025) [4]
Ω	$0.58950329 \dots$	$g = \mathbf{1}_{\text{prime powers}}$	Irrational (<i>ibid.</i> , remark)
φ	$1.36763080 \dots = S$	$g = A = \varphi * \mu$	Open (Erdős #249)

$\sum_{d \geq 1} \varphi(d)/(2^d - 1) = 2$ (checked) — and the signed squared-Mersenne lens $S = \frac{1}{2} + \sum_{d \geq 1} \mu(d)/(2^d - 1)^2$ (checked) recomputes S to all 50 places. The difficulty lives on the power-series side, and **R1** is a statement about exactly that side.

There is also an exact probabilistic reading, and it is more than a gloss. Let X and Y be independent fair-coin geometric waiting times on $\mathbb{N}_{>0}$. Then

$$S = \frac{1}{2} + \Pr[\gcd(X, Y) = 1]$$

(checked), the probability being the visible-coprime-pair series $\sum_{\gcd(m,n)=1} 2^{-(m+n)}$. That probability is distributed over the Stern–Brocot tree in a completely explicit way: the Mersenne weights of the positive reduced slopes sum to exactly 1 (checked), and at each node (a, b) the cylinder mass $1/((2^a - 1)(2^b - 1))$ splits exactly into its stop mass and the masses of its two children (checked). So $S - \frac{1}{2}$ is the total mass of an exactly self-similar subdivision of a probability space. What is missing is not structure; it is an arithmetic consequence of the structure.

3 The 2-kernel of Euler’s totient

3.1 What was already known, and what is different here

For an integer $k \geq 2$, the k -kernel of a sequence c is $\{n \mapsto c(k^j n + r) : j \geq 0, 0 \leq r < k^j\}$, and c is k -regular when the module generated by its k -kernel is finitely generated; both notions are due to Allouche and Shallit ([1], Definitions 1.1 and 2.1). Coons proved that φ is not k -regular for any $k \geq 2$ ([2], Theorem 3.2 of the published version and Theorem 3.3 of the preprint, which numbers its results on a single running counter). Bell and Smertnig reach a further negative conclusion, and by a different route rather than by strengthening that one: their Theorem 1.3 shows that a k -Mahler series with multiplicative coefficients has a k -regular coefficient sequence in an explicit closed form, and that the totient generating series is not k -Mahler for any $k \geq 2$ follows directly from the closed form [3].

Coons’s statement is about the union over all levels: no finite set of sections generates the rest. Bell and Smertnig’s is about a functional equation. **R1** and **R2** are complementary positive statements at each finite level, and are consistent with both precisely because the rank is unbounded in e . Knowing that a span is not finitely generated says nothing about its dimension at a given truncation, nor about which relations hold there;

what is added here is that the dimension through level e is exactly $2^e + 1$, that an explicit family is a basis, and that the two elementary reductions generate every relation. We found no prior source giving any of those three for φ ; that is the outcome of a search, not a proof of non-existence, and no priority is claimed.

3.2 The mechanism

The producer for **R2** is a separated evaluation minor, and the linear algebra at its centre is simpler than the construction that feeds it. Evaluate the sections at one point apiece to get a square matrix M , and suppose each column j admits a fixed power 2^{d_j} dividing every entry of that column such that, after dividing the column through by it, the diagonal entry is odd and every off-diagonal entry is even. The normalised matrix is then the identity modulo 2; its determinant is odd, hence nonzero, and the sections are independent ([checked](#)). Producing evaluation points with exactly that parity pattern is what the arithmetic input is for. The Chinese remainder theorem together with Dirichlet's theorem on primes in arithmetic progressions makes one affine channel prime, which fixes the two-adic depth of the diagonal entry, while every other channel receives a fresh prime divisor congruent to 1 modulo a large power of two, which supplies the one extra factor of two that the off-diagonal entries need ([checked](#)). Even residues never enter, because the second reduction has already carried them to odd-core channels ([checked](#)).

R1 then needs only two further steps, and both are short. Linear independence is a property of finite subfamilies, so the level- e producer, applied with e above the largest level occurring in a given finite subset, already gives independence of the whole infinite family $\mathcal{B}$. And the two reductions carry every remaining section onto a rational multiple of a member of $\mathcal{B}$, so the spans agree. That the assembly is short is the point: the level- e theorem was always the whole content, and stating its consequence as infinite-dimensionality understated it. The sharp form is a basis.

Remark. **R1** and **R2** are statements about φ , not about S ; they are true, and their proofs are complete, whether or not S is irrational. They are here because they are the exact form of “no finite amount of the totient's dyadic structure determines the rest” — the property that **R6** consumes for an eventually periodic weight and that A lacks.

The rationality side. Binary long division turns a hypothetical rational value into an integer carry recurrence: for coefficients $c(n) \leq n$, and in particular for $c = \varphi$, the series $\sum c(n)/2^n$ is rational exactly when a tempered integer carry orbit exists ([checked](#)). Transporting **R2** through that recurrence gives **R4**. The scope line there is worth repeating: this is an anti-compression barrier, and the complementary finite-rank upper bound, which is what would make it a proof, is not available.

4 The denominator bound

R3 is a Farey exclusion, and we say so before saying anything else about it. The argument is the classical mediant one. The Lean lemma commits the first 240 binary digits of the shifted series as a single residue over the totient carry window $(N, K) = (1, 240)$; a rational a/q can equal S only if it falls into a resulting bad interval of width $243/2^{240}$; that interval is bracketed by two explicit unimodular fractions $a_1/b < c_1/d$ with $bc_1 - a_1d =$

1; and the mediant lemma (checked) forces any rational strictly between unimodular neighbours to have denominator at least $b + d$. The excluded range is therefore $q \leq b + d - 1$, which is the printed constant (checked), and the transport to the series is a tail estimate (checked). *The method is standard, and we claim no novelty for it.*

Two things should be read off correctly. First, the scale. A window of K computed binary digits yields a mediant bound of order $2^{K/2}$: the same pipeline gives 2.49×10^{17} at $K = 120$ (checked) and 7.96×10^{34} at $K = 240$. The constant is a function of how many digits were committed, not a measure of how much the problem has moved. Second, the sharpness. Within its window the bound cannot be improved at all: $b + d$ itself fails the certificate, so $b + d - 1$ is exactly the last denominator excluded (R3). A reader who regards an explicit Farey exclusion as routine is not in disagreement with this note. R1, not R3, is where we would ask a sceptical reader to look.

The bound transfers to the coprimality form of the constant with the constant halved. Writing the visible-coprime-pair series as $\sum_{\gcd(m,n)=1, m,n \geq 1} 2^{-(m+n)}$, one has $S = \frac{1}{2} + \sum_{\gcd(m,n)=1} 2^{-(m+n)}$ (checked), so that series has no representation a/d with $d \leq 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526$ (checked), that number being $(q_0 - 1)/2$ for the constant q_0 of R3. This series is the probability that two independent geometric fair-coin waiting times are coprime; the checked statements are the series identity and the series exclusion, and the probabilistic reading is a gloss on them rather than a formalised measure-theoretic statement.

5 Barriers, and two we do not state as theorems

R7 is the one method barrier this note states, and its scope lines are given with it in Section 1. It is worth a sentence of mechanism, because the mechanism is the whole of it: take a prime $p > \max(D, 2)$; then $A(p) = p - 2$ by R5, and $p \nmid D(p - 2)$, so $D \cdot A(p)/p \notin \mathbb{Z}$. Being one line is a virtue here. The statement is about the weight the abstract already names, it needs no vocabulary this note has not defined, and it survives a hostile reading.

Two further no-go results in the formal corpus appeared as propositions in an earlier version of this note, and are demoted rather than dropped, because the reason is more useful to a reader than the propositions were.

A fixed-precision transport result (checked) says that at every fixed positive precision, each finite compatible valuation-unit word admits a prefix-locked centred completion. Its statement quantifies over abstract pairs of a two-adic valuation and an odd unit; it mentions neither φ nor S , its content is that a congruence class meets any interval of the corresponding length, and it is a restatement of a lemma printed immediately above it. The class of arguments it closes — deriving a contradiction from a local signature read at a precision fixed in advance, along a finite word — is not one anybody has proposed for #249.

A factor-ideal result (checked, with a sparse-anchor companion at checked) exhibits, for each $t \geq 3$ with $H = \text{lcm}(1, \dots, t)$, a nonzero integer carry whose forcing letters lie in the ideal generated by $\varphi(H)$, which reproduces the true totient differences at $t - 2$ prescribed indices, and which every finite integer shift polynomial carries to a pair with the same coboundary form, the same ideal memberships and an ℓ^1 -weight bound. It

does close that hypothesis class. Two limits. The witness is a spike — its state is $-\varphi(H)$ at $t - 2$ points and zero elsewhere, so it stays uniformly bounded while the strip bounds it respects grow, and the construction is cheap for that reason. And the shifted pairs are not asserted to be nonzero, so an argument that additionally demanded non-vanishing after the shift is not closed. An earlier draft of this note also attributed to it a whole-ray anchor condition, a diagonal-bound condition and a strict-survivor condition that the cited theorem does not contain, and described it as “surviving” the shift algebra when the checked conclusion carries no non-vanishing clause. Correcting that is the reason it is here at all.

6 Three problem-owned programmes

The modules below are problem-owned rather than part of the reviewed corpus. Each states its unproved hypothesis explicitly; none advances #249, and we list them because a reader deciding whether to spend time here should see the sufficient conditions in the exact form Lean checks them.

Strict prime-orbit escape. The cleanest of the three, because it moves a threshold. The producer [asks](#) for cofinally many primes p at which the first tail-orbit exponential has real part strictly below $9/10$. Granting it, [a positive adaptive truncation budget](#) follows, and then [S is irrational](#). The point of interest is the constant: the legacy route required a $4/5$ gap, this one requires only $9/10$, so the sufficient condition is strictly weaker while the conclusion is unchanged. Weakening a hypothesis you still cannot verify is progress on the *statement*, not on the problem, and the producer remains unproved. Section [7](#) states the sharper form in which this hypothesis is now available.

Finite Euler-sieve algebra. Elementary identities behind the squared-Möbius approximation to the totient generating function; no transcendence input is claimed. The local Euler factors are exact squares at $s = 1$ and $s = 2$, and on the divisor-sum prime-power row [the first difference](#) is $p - 1$ while [the second difference](#) is $p^{e+1}(p - 1)$. The second difference is the content: applying $\mu * \mu$ converts the divisor-sum row into the prime-power totient row exactly, at every finite stage. These are one-line facts; they are recorded because the approximation they license is otherwise easy to assert without checking.

Prime-ray cyclotomic curvature. A producer/consumer interface with two named obligations. Its curvature functional is [the checkerboard](#) on a two-anchor rectangle, which [annihilates every separable background](#) $u(x) + v(y)$ and is [the unique such \$2 \times 2\$ annihilator up to scale](#). That uniqueness is worth stating: the operator is not a convenient choice among many, it is the only one, which is what makes the resulting obstruction coordinate-free. On top of it sit [a squarefree divisor-layer cancellation](#) and [a centred-completion step for residue classes modulo an even modulus](#). The two obligations are [eventual clean cyclotomic layers on a prime ray](#) and [an exact-order witness at every rational prime divisor of a layer](#); together they give [escape from every prescribed finite prime support](#). Neither obligation is proved here, and polynomial resultant realisability and Archimedean growth remain upstream of both. Obligation (b) is a primitive-divisor question in classical shape: it asks that every prime divisor of a layer $C(mq)$ have mul-

multiplicative order at most d relative to the ray index mq . One line, offered as prose and not as a checked statement: (b) alone already forces the escape by size, since $p \mid C(mq)$ gives $mq \leq p^d - 1$, so for a finite set S of primes no member of S divides $C(mq)$ once $q > \max_{p \in S} p^d / m$; the role of (a) is then to keep the conclusion from being vacuous, since $C(mq) = 1$ would satisfy it trivially. Whether (b) is known, in either direction, for cyclic resultants of an integer polynomial is the question we would put to a specialist first.

7 The exact frontier

The strongest reduction in this work is worth stating in the sharpest form the corpus now supports rather than the form an earlier draft used.

The obligation. Write $H_t = \text{lcm}(1, \dots, t)$ ([definition](#)) and let $R_N = \sum_{m \geq 1} \varphi(N+m)/2^m$ be the fractional layer of $2^N S$ ([definition](#)). *Wanted:* for every t_0 there is a $t \geq t_0$ with $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$. By [R8](#) that supply is not merely sufficient; it is equivalent to #249. Nothing short of a cofinal statement is therefore of use, and no finite band approximates one. The converse direction is sharper than it looks: one pair $h > 0, N$ with $R_{N+h} - R_N \in \mathbb{Z}$ already makes S rational ([checked](#)).

The certificate is decidable at each t and has been kernel-checked to fire at the 28 scales ([listed](#)) lying between $t = 1$ and $t = 64$ ([checked](#)). Later work in the repository extends the checked band to every $t \leq 82$; the module proving it postdates this note's pinned snapshot and so is not linked here, and the claim registry records the wider band. The distinction does not matter for the obligation: a band bounded at $t = 64$ contributes nothing at $t = 65$, and a band bounded at $t = 82$ contributes nothing at $t = 83$.

An analytic route, with the budgets made explicit. The most concrete approach the corpus supplies replaces the qualitative prime-orbit gap of Section 6 by a decomposition with numbers in it. A constant-saving first-harmonic bound on one dyadic block, or on any nonempty finite set of indices, already forces a certificate ([checked](#)). The block sum in question decomposes exactly into four named terms — a centred supplier-fibre correlation, a fibre-mean contribution, a bad-set contribution and a non-supplier contribution ([checked](#)) — and meeting four explicit numeric budgets on those terms cofinally already gives irrationality ([checked](#)). So the analytic obligation is not “prove a cancellation estimate” in the abstract: it is to beat four stated constants on a decomposition that is already exact.

Not an ask. The constant 7.96×10^{34} of [R3](#) is not an open item. It is the classical Farey mediant bound applied to a window with a free parameter K , it has order $2^{K/2}$, and raising it needs only more committed binary digits — a computation, not an idea.

8 What we need from a mathematician

Three obligations carry the whole conditional part of this work. Each is a statement about one explicitly defined sequence, each is stated below from definitions given here, and none needs the rest of this note — or any Lean — to be understood, proved or refuted. They are written in the form in which we would want an answer.

Two remarks apply to all three. A negative answer is worth exactly as much to us as a positive one: if a route provably cannot supply what it is asked for, we will say so here and drop the route rather than leave it listed as live. And none of the three asks anyone to grant anything about S ; each is an ordinary question about a defined object, answerable, refutable or recognisable from the literature on its own terms.

Ask 1. Certificates beyond every cutoff, on the lcm diagonal. Write $H_t = \text{lcm}(1, \dots, t)$ (definition) and let $R_N = \sum_{m \geq 1} \varphi(N+m)/2^m$ be the fractional layer of $2^N S$ (definition). *Wanted:* for every t_0 there is a $t \geq t_0$ with $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$. That is the hypothesis on the right of the diagonal normal form, in the finite-certificate coordinates the two forms share (checked).

Already settled, so that nobody repeats it. The normal form is an equivalence, not a sufficient condition: S is irrational if and only if that supply exists. Nothing short of a cofinal statement is therefore of use, and no finite band approximates one. The converse direction is sharper than it looks — one pair $h > 0, N$ with $R_{N+h} - R_N \in \mathbb{Z}$ already makes S rational (checked). The certificate is decidable at each t , and has been kernel-checked to fire at the 28 scales (listed) lying between $t = 1$ and $t = 64$ (checked). Lengthening that list is not progress towards the ask, and is not offered as such: a band bounded at $t = 64$ contributes nothing at $t = 65$.

Most useful first. (i) Any argument giving $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$ for infinitely many t , with no uniformity or rate required. (ii) A reason the lcm diagonal is the wrong ray to ask this of — for instance that $R_{2H_t} - R_{H_t}$ is beyond the reach of present methods for structural reasons, not merely unproved. (iii) A single t with $R_{2H_t} - R_{H_t} \in \mathbb{Z}$. A construction or a proof (i) helps most; (ii) is next and would be acted on.

If answered. (i) closes #249, since the normal form is an equivalence and the deduction is already checked. (ii) retires the diagonal presentation, and with it Ask 1, without touching the problem. (iii) settles #249 in the opposite direction, which would be a larger result than anything asked for here.

Ask 2. One cofinal prime with first-harmonic real part below $9/10$. *Wanted:* for every $h \geq 1$ and every N_0 , a prime $p \geq \max(N_0 + h + 1, h + 5)$ with

$$\cos(2\pi (R_{p-1} - R_{p-h-1})) < \frac{9}{10},$$

equivalently with $R_{p-1} - R_{p-h-1}$ further than $\arccos(9/10)/2\pi = 0.07178\dots$ from the nearest integer. Written with $N = p - h - 1$, this is the hypothesis verbatim: the real part of the first additive character of $R_{N+h} - R_N$, below $9/10$ for cofinally many primes.

Already settled. The quantity is not an artefact of the truncation: integer prefixes cancel under the character, so this is exactly the first additive character of $2^N(2^h - 1)S$ (checked), that is, of one orbit of the doubling map. Granting the gap, the adaptive truncation budget (checked) and then irrationality (checked) follow with no further input. The threshold has already been moved: the legacy route consumed a $4/5$ gap and this one consumes $9/10$, so the hypothesis is strictly weaker than the one it replaced. We have no proof of it at either threshold.

Most useful first. (i) A reference. This is a one-sided escape statement for a single explicit real orbit sampled along the primes, and an analytic number theorist may recognise it as standard, as a known consequence of an existing estimate, or as a known hard

case; any of the three would be decisive for how we spend time. (ii) A proof or proof sketch. (iii) A reason the statement is out of reach for orbits of this kind. (i) helps most, because the cost of the wrong answer here is measured in months.

If answered. (i) or (ii) closes #249 through the checked endpoint above. (iii) closes only this branch: the gap is sufficient for irrationality and nothing here shows it necessary, so its failure would leave the problem exactly where it is. This is the ask we consider most likely to be answerable by somebody who has not read the rest of this note.

Ask 3. Bounded multiplicative order along a prime ray. Fix integers $m, d \geq 1$ and a layer sequence $C: \mathbb{N} \rightarrow \mathbb{N}$. Two statements are wanted, of which the second is the substantial one. (a) There is a Q_0 with $C(mq) > 1$ and $\gcd(C(mq), mq) = 1$ for every prime $q \geq Q_0$ ([the supply](#)). (b) Whenever q and p are prime and $p \mid C(mq)$, some k with $1 \leq k \leq d$ has $mq \mid p^k - 1$ ([the consumer](#)). Statement (b) is a primitive-divisor question in classical shape: it asks that every prime divisor of a layer have multiplicative order bounded by d relative to the ray index mq .

Already settled. The curvature coordinates are forced rather than chosen — the checkerboard is the only 2×2 joint annihilator up to scale ([checked](#)) — and the divisor-layer cancellation ([checked](#)) and the affine recentring step ([checked](#)) are proved from their stated hypotheses. What is *not* there should be said plainly: (a), (b) and the intended consequence [finite-prime-support escape](#) are all recorded as definitions, and the module proves neither obligation. One line, offered as prose and not as a checked statement: (b) alone already forces the escape by size, since $p \mid C(mq)$ gives $mq \leq p^d - 1$, so for a finite set S of primes no member of S divides $C(mq)$ once $q > \max_{p \in S} p^d / m$. The role of (a) is then to keep the conclusion from being vacuous, since $C(mq) = 1$ would satisfy it trivially.

Most useful first. (i) A reference: whether (b) is known, in either direction, for cyclic resultants of an integer polynomial, which is the intended source of C . (ii) A counterexample family for which (b) fails. (iii) A construction of a layer sequence satisfying both. A reference helps most, because we do not know whether we are asking something classical or something false.

If answered. This closes one branch and not the problem. Escape from every prescribed finite prime support is not irrationality of S : realising C by cyclic resultants at all, and the Archimedean growth estimate, are both upstream, both unproved, and no route from escape to #249 is claimed here. An answer tells us whether to keep the branch; it does not move #249.

Not an ask: the denominator bound. The constant 7.96×10^{34} of [R3](#) is not an open item, and help with it is not wanted. It is the classical Farey median bound applied to a window with a free parameter K , it has order $2^{K/2}$, and raising it needs only more committed binary digits — a computation, not an idea. A reader whose first instinct is to improve that number would be spending effort at the one point in this note where the method is completely understood and the return is nil.

9 What remains open

The problem is open and this note does not narrow it. Precisely:

1. Irrationality of S (R10). No proof is claimed.
2. The certificate routes of R8 convert irrationality into the existence of an unbounded supply of finite tail-difference certificates. Finite instances are checked — the 28 diagonal scales of Section 7 — and the unbounded supply is not. *An equivalence is not progress.* R3 is a finite instance of a *different* family, the gap certificates of R9, which reach irrationality by a one-way implication; the two families should not be counted together.
3. The four pivot budgets of Section 7, and the prime-orbit gap they refine, are unproved.

The correlation source that fits the normalised totient most directly is Balasubramanian–Giri–Srivastav [5], not an off-the-shelf transfer of the Tao–Teräväinen engine. Write

$$g(n) = \frac{\varphi(n)}{n} = \sum_{d|n} \frac{\mu(d)}{d} = (f * 1)(n), \quad f(d) = \frac{\mu(d)}{d} \in \mathcal{A}_1.$$

Theorem 2 of the arXiv version of [5] gives, uniformly for $|h| \leq x/2$, an explicit asymptotic for $\sum g(n)g(n-h)$ with error $O(\log^2 x)$; Remark 4 gives its Euler product, and the weighted partial-summation formula immediately following Corollary 8 permits $Q(n) = n(n-h)$, restoring the two linear factors needed to return from $g(n)g(n-h)$ to $\varphi(n)\varphi(n-h)$ with an explicit propagated error.

This corrects a tempting but inaccurate description in an earlier draft. Tao–Teräväinen’s quantitative correlation theorem ([4], Theorem 10) assumes either quantitative equidistribution together with an exact small-prime condition, or non-pretentiousness. Here $g(p) = 1 - 1/p$, so the stated small-prime condition does not hold, while $\sum_p (1 - g(p))/p = \sum_p 1/p^2 < \infty$, so g is pretentious to the constant function. Their theorem therefore does not apply unchanged. The Balasubramanian–Giri–Srivastav theorem supplies the missing first- and second-moment input in the correct divisor-convolution class, but it does *not* supply the residue small-ball or phase anti-concentration estimate needed to turn those moments into a certificate. That modulo- 2^L step is the precise remaining analytic gap; no irrationality conclusion is drawn from the correlation asymptotic alone.

9.1 Unregistered results

Two of the items above are checked propositions that no row of the claim registry currently owns: the three declarations of R1, and the five of R5; the supply implication of R9 is likewise unowned, and R3 cites three declarations of which the registry’s row owns one. Under this project’s own division of authority the registry, not a manuscript, owns public status, so the honest tag on those items is that they are kernel-checked and not yet registered. We print that rather than borrowing a neighbouring row’s status. The gap is a defect in the record, not in the proofs, and closing it is a review action rather than a mathematical one.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions, and kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. The consequence drawn in the last sentence of [R1](#), the deduction in [R6](#) that an unbounded weight is not eventually periodic, the mechanism sentence in Section 5, and the size argument in Section 6 are one-line arguments in the prose, marked as such; the first three are drawn from checked statements, and the last is drawn from two hypotheses that are not proved. Results attributed to Allouche and Shallit, to Coons, to Bell and Smertnig, to Luca and Tachiya, and to Tao and Teräväinen are cited from the literature and are not formalised here; the assessment that no prior source gives an explicit basis, an exact finite-level dimension formula, or a relation normal form for the totient 2-kernel is the outcome of a literature search and is not a proof of non-existence.

Scope of the AI declaration. The note on the first page states that the prose was model-generated. The scope is wider than that sentence alone conveys, and the wider statement is the accurate one: large-language-model agents assisted throughout, drafting and revising this exposition, the Lean developments it links to, and the repository software that checks them. Will Cook set the objectives, selected and reviewed the public claims, checked the cited literature, and authorised this manuscript, and is responsible for its contents. Formal authority is the pinned kernel's acceptance of an exact proposition; no model output carries any, and neither does this sentence.

Erdős Problem #249 remains open.

References

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- [2] M. Coons, *(Non)Automaticity of number theoretic functions*, J. Théor. Nombres Bordeaux 22 (2010), no. 2, 339–352; [arXiv:0810.3709](#). Theorem 3.2 in the published version (Theorem 3.3 in the preprint): φ is not k -regular for any $k \geq 2$.
- [3] J. Bell and D. Smertnig, *Mahler series with multiplicative coefficient sequences*, 2026, [arXiv:2603.23456](#). The totient generating series is not k -Mahler for any $k \geq 2$, a consequence of their Theorem 1.3 recorded in the introduction.
- [4] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, [arXiv:2512.01739](#) (submitted December 2025, revised April 2026). Theorem 1.3 proves irrationality of $\sum_{n \geq 1} \omega(n)/2^n$ at base 2; the extension to every integer base and the Ω analogue are stated as remarks, with the modifications left to the reader.
- [5] R. Balasubramanian, S. Giri and K. R. Srivastav, *On correlations of certain multiplicative functions*, J. Number Theory 174 (2017), 221–238, [DOI](#); [arXiv:1511.02221](#). The arXiv version's Theorem 2 and Remark 4 give the uniform shifted divisor-convolution correlation and Euler product used above.

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- [7] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, 1980, p. 61.
- [8] T. Bloom, *Erdős Problems*, <https://www.erdosproblems.com/249>.